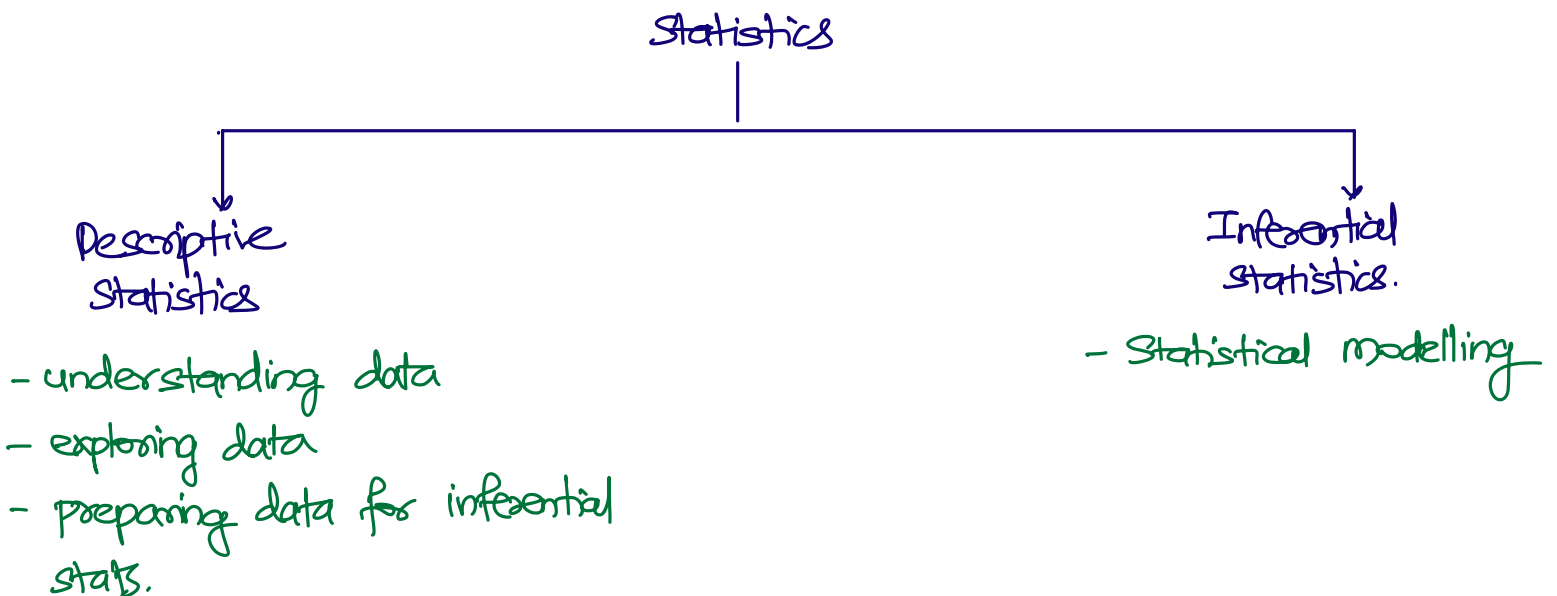
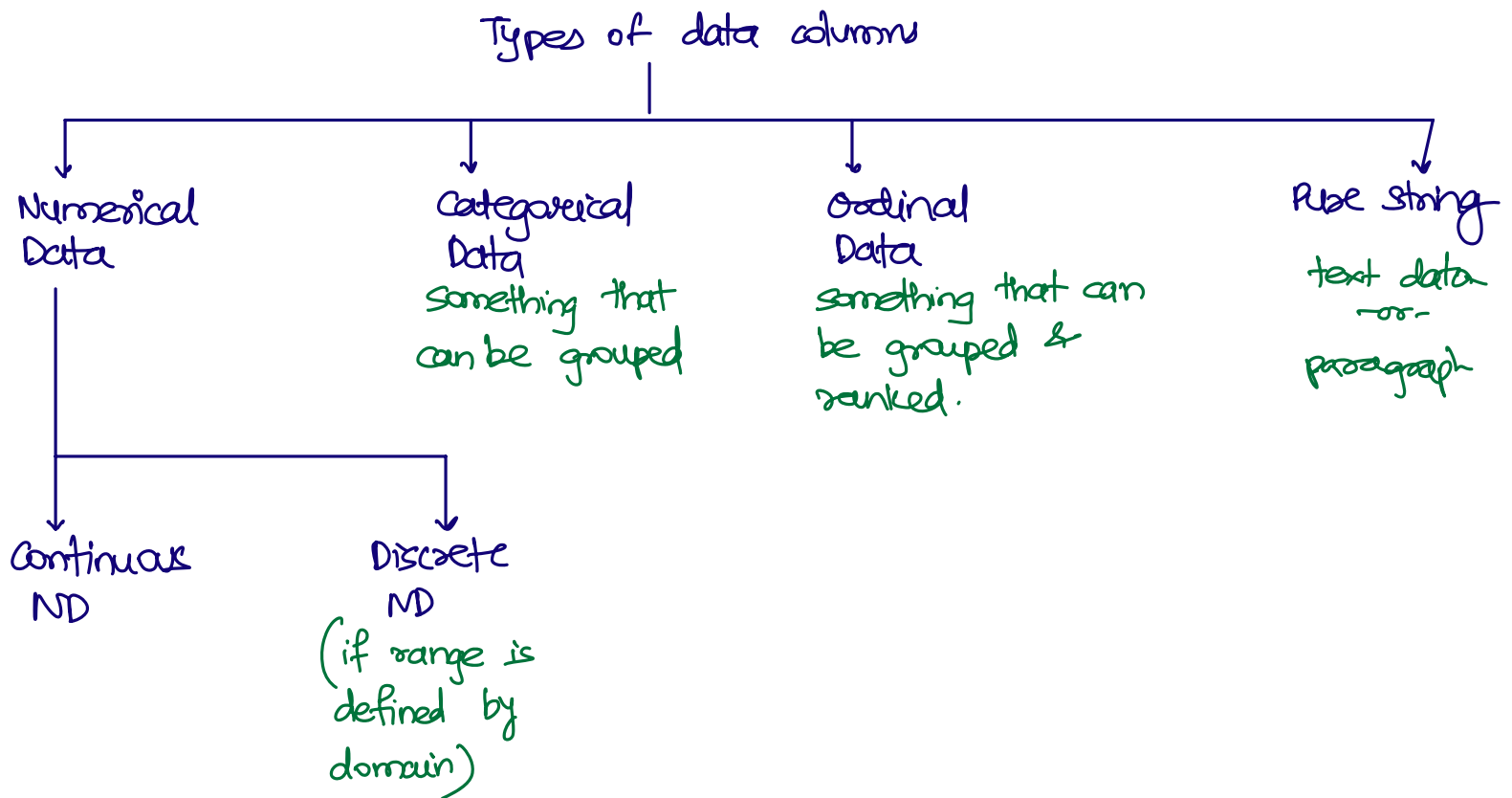




Descriptive Statistics

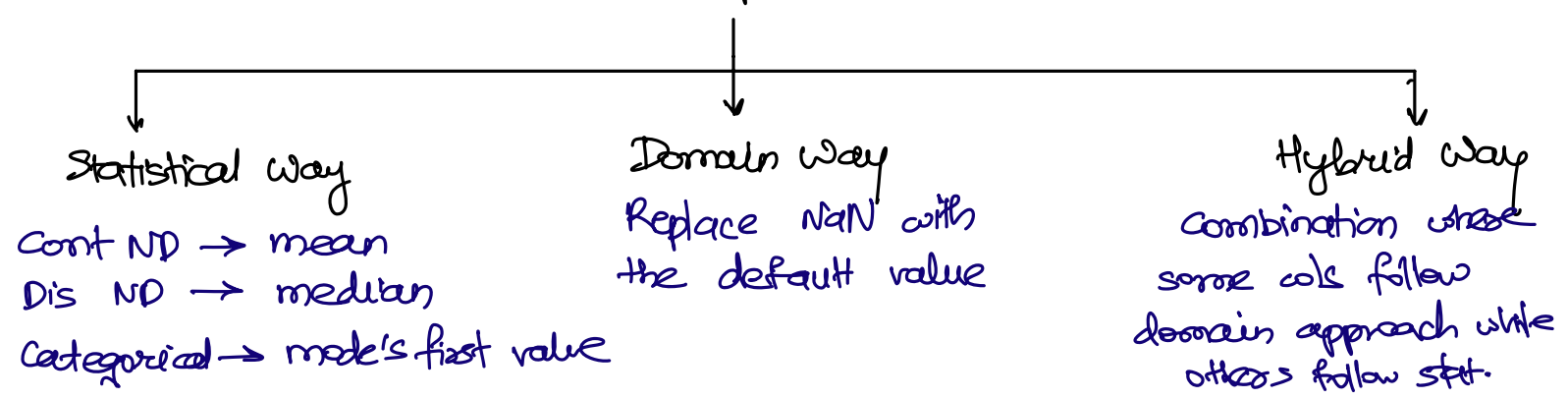
Data Exploration Phase -or- Exploratory Data Analysis (EDA)

Descriptive Stats is the first step to deal with the SAMPLE. Typical operations that a data analyst, data scientist, business analyst prefer:

normal ←-----→ skew

- ① Check the type of **DISTRIBUTION** for each numerical column.
 - ② Identify and deal with **INAPPROPRIATE** data.
 - ③ Identify and deal with **OUTLIERS**
 - Tukey's method (1.5 IQR Rule) (box plot)
 - Z-score
 - ④ Identify and deal with **CATEGORICAL** data.
 - OHE (One Hot Encoding) Imputation. (Dummy Variables)
 - ⑤ Identify and deal with **MISSING VALUES**
 - Statistical way
 - Domain way
 - Hybrid way
 - ⑥ Identify and deal with **ORDINAL** data.
 - Replace category with its associated rank as decided by domain experts
 - ⑦ Discover any
 - corr analysis
 - skewness
 - kurtosis
 - feature elimination
- ASSOCIATION**
RELATIONSHIP
PATTERN
- Hypothesis Testing → $H_0 \rightarrow \text{sad face}$
→ Statistical Testing → $H_a \rightarrow \text{happy face}$
→ confirm who wins
- between two or more VARIABLES (columns) in the given dataset.

Missing Value Analysis (Imputation)

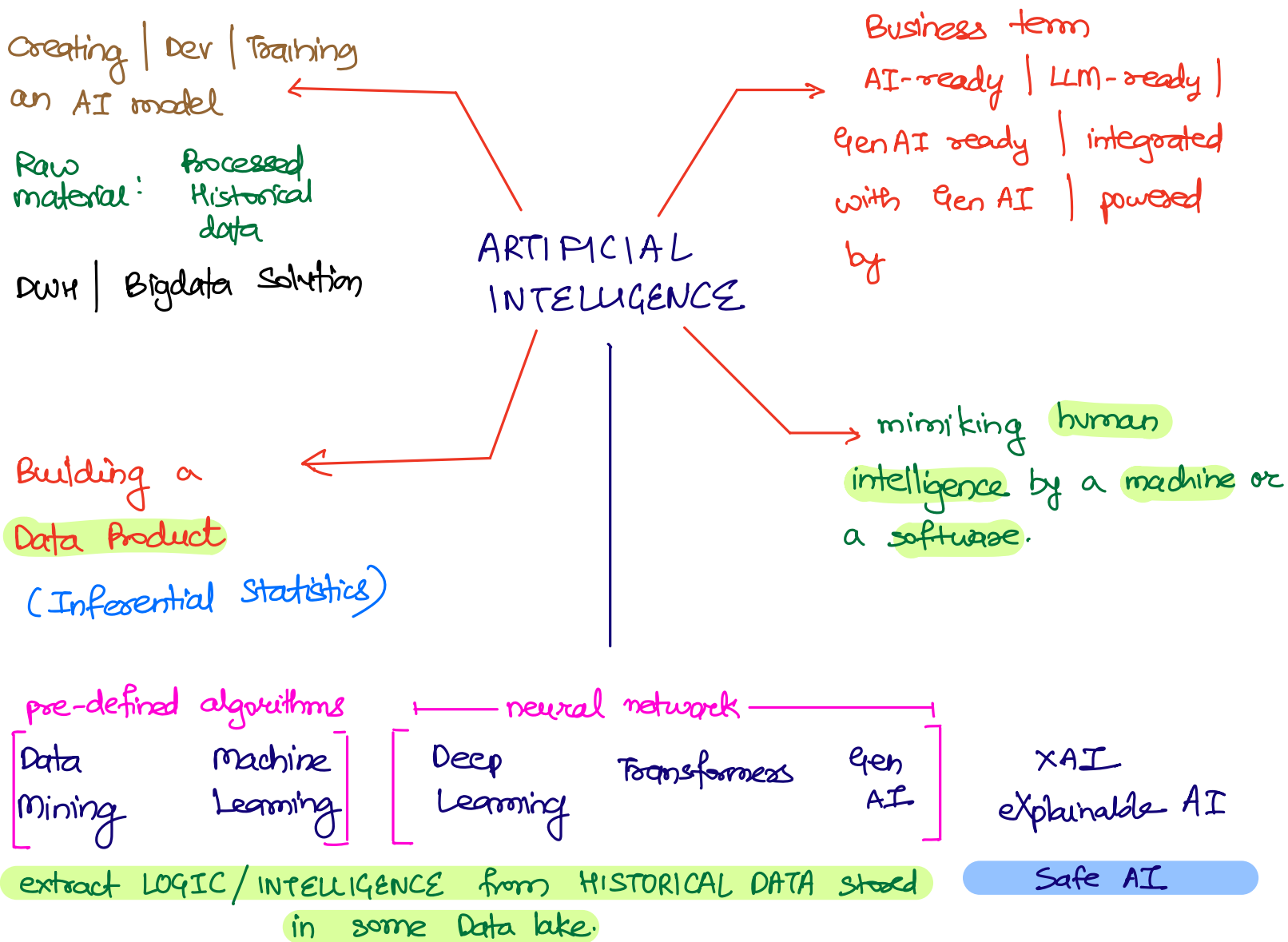


SL (Significance level) → reveals how much error in prediction or analysis the business can tolerate.

Standard α values → 0.05, 0.01, 0.1

CL (Confidence level) → how much minimum accuracy to be achieved by the data product.

$$\underline{CL = 1 - SL}$$



creating an
Artificial
Intelligence
product

} → Data Product

modulare Programming
Practices

Automated Learning
Practices.

Here the programmer must be
aware of the LOGIC to
reach the solution.

Here the programmer will make use
of HISTORICAL DATA to DERIVE LOGIC
to reach the solution

SQL engg → DWH
[Preserve
historical
data]

DM
[prepare models
out of the data]

Bigdata
engg & → Bigdata
statistical
experts frameworks
eg. Hadoop
spark

ML
Apache mahout

Bigdata
engg & → Data
AI engg Lake

ML, DL
eg. python frameworks
sklearn,
tensorflow keras,
pytorch.

* AI engg → Data
(Generative
models) Lake
(unstructured
data)

Generative model

Machine Learning (it uses data mining algo to derive a model out of historical data)

Supervised Learning

Given dataset have both features and label
(questions) (answers)
(independent variables) (dependent variables)

Given dataset will have question and answer. The goal of supervised learning is to discover the logic to derive answer.

Unsupervised Learning

Given dataset will have only features

Goal: Extract hidden patterns, Extract hidden groups, associations.

Supervised learning algorithms

Regression algorithms

Label: Continuous ND

Classification algorithm

Label: Categorical Data
or
Discrete ND.

Model Building process in AI

Assuming the given dataset is compatible for Inferential stats.

- ① Data must be COMPLETE.
- ② Data must be STRICTLY NUMERIC.

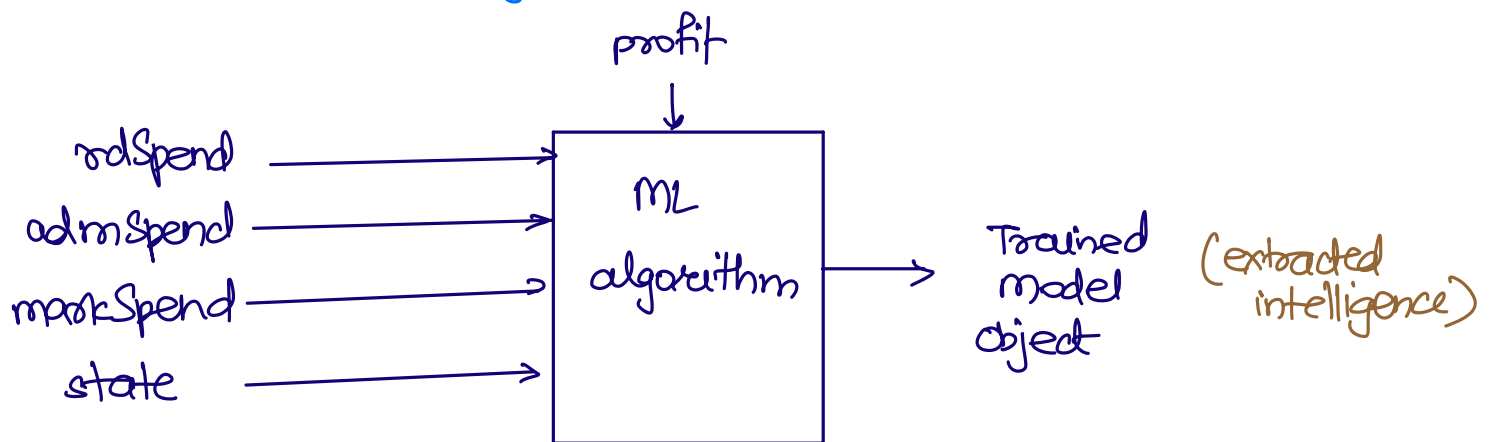
① Separate data as features and label (applicable for supervised learning problems)

② Split the data into two sets:

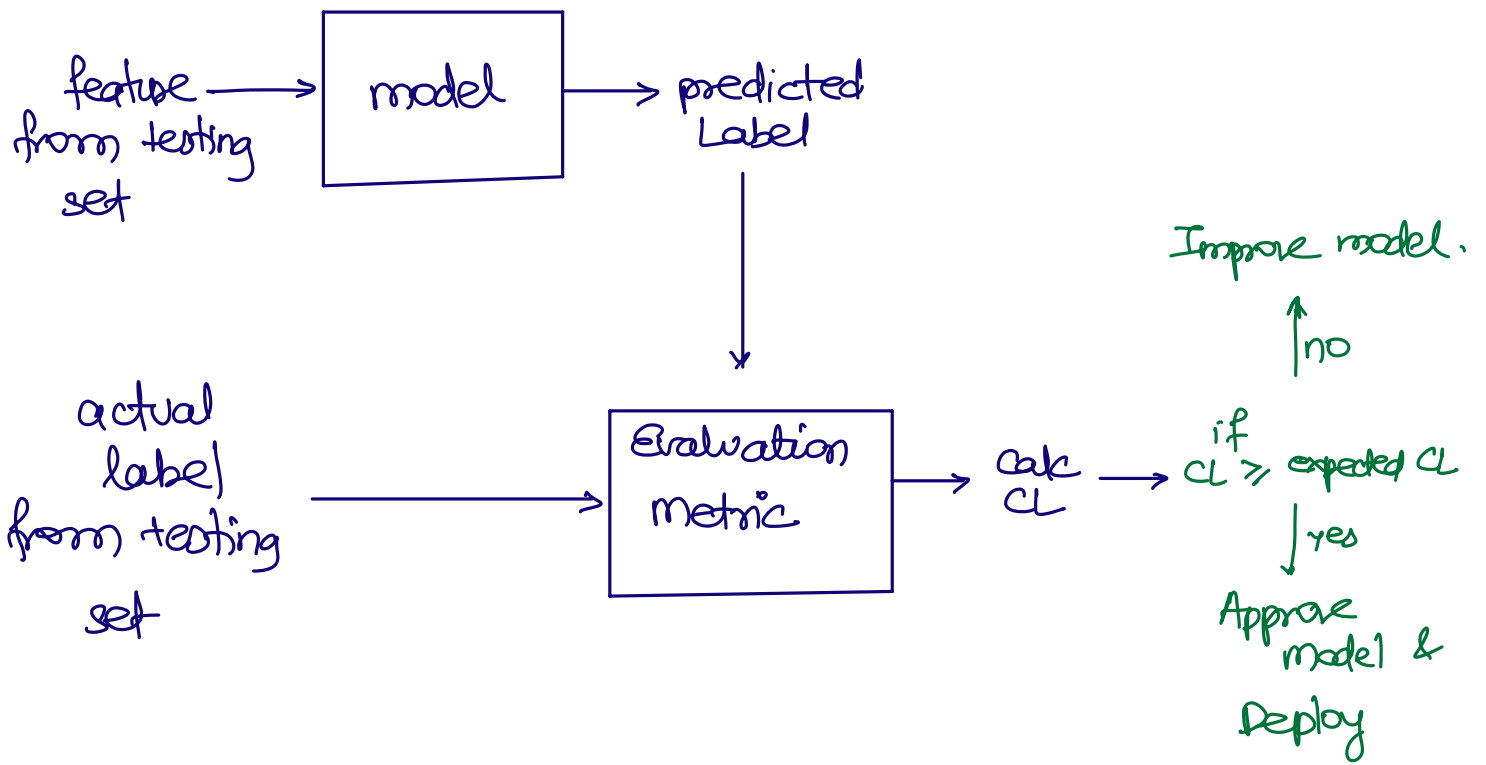
- * Training Set → used to TRAIN the model
- * Testing Set → used to EVALUATE the model
(check the quality of predictions)

③ Train the model using training set and appropriate ML algo.

eg: Create a model that can predict the profit of the company based on company's spending pattern and company's location.



④ Check the quality of the model. | model Evaluation



⑤ Deploy the model